

Business Requirement Document (BRD)

Logistics ERP Analytics

1. Introduction

This document defines the business requirements for developing a Logistics ERP Analytics solution using the provided dataset. The goal is to enable data-driven decision-making across supply chain, inventory, procurement, and logistics operations.

2. Business Objectives

- Improve delivery performance and reduce delays
- Optimize inventory levels (avoid stockouts and overstock)
- Enhance supplier performance monitoring
- Improve operational efficiency and cost control
- Enable executive-level reporting and insights

3. Stakeholders

- CEO: Strategic decision-making
- CFO: Financial performance monitoring
- COO: Operational efficiency
- Warehouse Manager: Inventory and stock control
- Procurement Manager: Supplier performance
- Logistics Manager: Delivery and shipment tracking

4. Data Overview

The dataset contains 65 columns including customer, product, supplier, warehouse, order, shipment, and inventory data. Key data domains include customer details, product information, order transactions, shipment tracking, and inventory movement.

5. Key Business Processes

- Order Management (Order ID, Order Date, Order Qty)
- Inventory Management (Opening Qty, Closing Qty)
- Procurement (Supplier, GRN, Received Qty)
- Logistics (Shipment ID, Courier, Delivery Dates)
- Warehouse Operations (Stock movement, adjustments)

6. Functional Requirements

- Dashboard creation for all business roles
- KPI tracking (Delivery Time, Inventory Turnover, Supplier Reliability)
- Drill-down and filtering capabilities
- Real-time and scheduled data refresh
- Predictive analytics for forecasting

7. KPIs

- Average Delivery Time
- On-Time Delivery Rate
- Inventory Turnover Ratio
- Stock-out Rate
- Supplier Lead Time
- Order Fulfillment Rate

8. Non-Functional Requirements

- Performance: Fast dashboard loading
- Security: Role-based access (RLS)
- Scalability: Handle large datasets
- Reliability: Error logging and monitoring

9. Assumptions & Constraints

- Data is clean and consistent
- Power BI will be used for visualization
- SQL/Python used for backend processing

10. Success Criteria

- Improved decision-making speed
- Reduction in delivery delays
- Optimized inventory levels
- Increased operational efficiency